

INTERNSHIP PROJECT

Analysis of India's Export Performance

FY 2015–16 to FY 2025–26

Project Overview

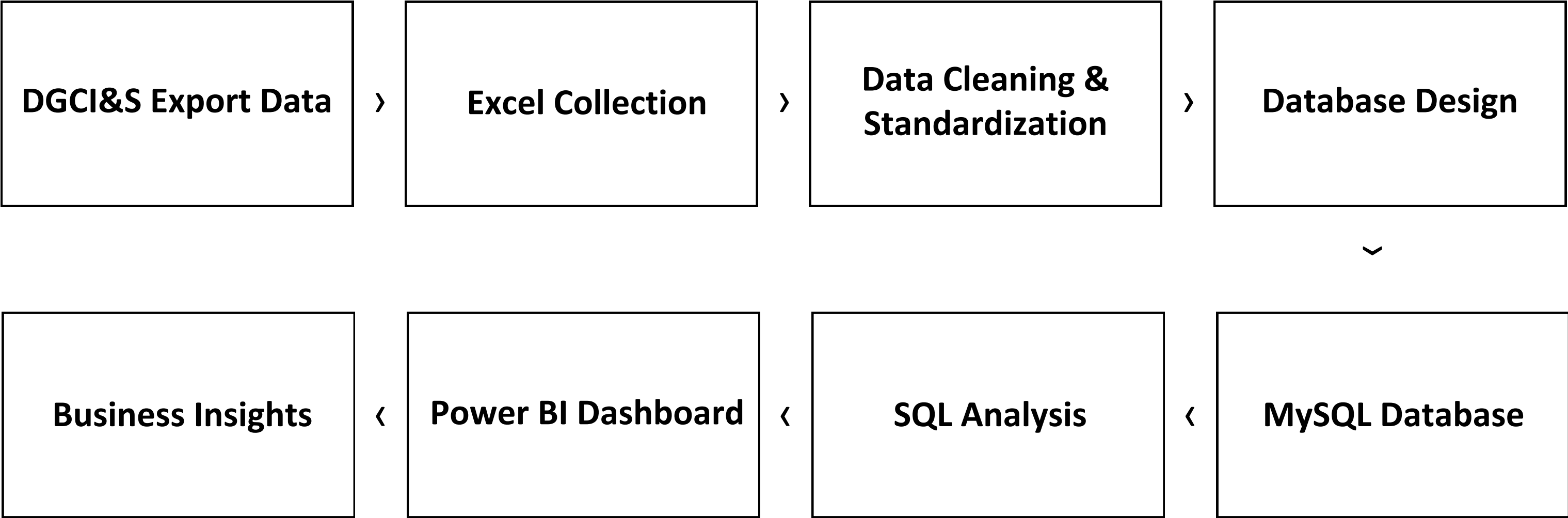
The objective of this project is to analyze India's export performance over the last ten financial years using official DGCI&S export data.

- Build a centralized export database
- Perform analytical SQL queries
- Build interactive Power BI dashboards
- Generate business insights from export data
- Collecting information about Compliance at Destination
- Listing all available modes of Transport and primary export routes

Technical Skills & Tools Used

Category	Tool / Technology
Database	MySQL
Data Processing	Microsoft Excel
Data Visualization	Power BI
Query Language	SQL

Project Workflow



Each stage feeds directly into the next — from raw DGCI&S data through to trade intelligence outputs.

Data Collection

Directorate General of Commercial Intelligence and Statistics (DGCI&S)

- Export data for FY2015–16 to FY2025–26
- Multiple Excel sheets for every financial year
- Commodity-wise export reports
- Country-wise export reports
- State-wise export reports
- Principal Commodity Code based commodity data



Government of India
Ministry of Commerce and Industry
Directorate General of Commercial Intelligence and Statistics
(An ISO 9001:2015 Certified Organisation)



Data Query By Principal Commodity

From :

From Date

To :

To Date

Export : ☐ Import : ☐

US \$: ☐ INR : ☐ Both : ☐

Report Type :

Sorted By :



Data Preparation & Standardization

- Examined structure of every Excel workbook
- Standardized all column names across yearly files
- Unified yearly datasets into a common schema
- Standardized country names for consistency
- Created country-to-continent mapping
- Prepared master dataset for analysis

Financial_Year	Commodity	Country_of_Destination	Unit	Quantity	Value_USD
2016	TEA	AFGHANISTAN	KGS	1030090	1819194
2016	TEA	ALGERIA	KGS	74542	143486
2016	TEA	ANGOLA	KGS	360	2880
2016	TEA	ARGENTINA	KGS	210	8006
2016	TEA	ARMENIA	KGS	1677	8733
2016	TEA	AUSTRALIA	KGS	3421182	20448926
2016	TEA	AUSTRIA	KGS	2032	15440
2016	TEA	AZERBAIJAN	KGS	181795	399818
2016	TEA	BAHARAIN IS	KGS	396324	1786924
2016	TEA	BANGLADESH PR	KGS	9136173	12301722
2016	TEA	BELARUS	KGS	10	30
2016	TEA	BELGIUM	KGS	573189	1225939
2016	TEA	BENIN	KGS	257304	1291081
2016	TEA	BHUTAN	KGS	13098	79267
2016	TEA	BOSNIA-HRZGOVIN	KGS	324	883
2016	TEA	BOTSWANA	KGS	1484	8365
2016	TEA	BRAZIL	KGS	539	6273
2016	TEA	BRUNEI	KGS	628	2090
2016	TEA	BULGARIA	KGS	192	723
2016	TEA	BURKINA FASO	KGS	225881	1463673
2016	TEA	CAMBODIA	KGS	755	3954

Database Design

Database Platform: MySQL

Master Table — Key Columns

- Financial Year
- Commodity Code
- Commodity
- Country
- Continent
- Unit
- Quantity
- Trade Value (USD)

Financial_Year	Commodity	Country	Unit	Quantity	Trade_Value_USD	PC_Code	Continent
2025	TEA	AFGHANISTAN	KGS	250798.00	612030	A1	Asia
2025	TEA	ALBANIA	KGS	431766.00	1913560	A1	Europe
2025	TEA	ALGERIA	KGS	212451.00	336576	A1	Africa
2025	TEA	ANGOLA	KGS	299649.00	1662099	A1	Africa
2025	TEA	ARGENTINA	KGS	26452.00	61609	A1	South America
2025	TEA	ARMENIA	KGS	20389.00	54919	A1	Asia
2025	TEA	AUSTRALIA	KGS	2990517.00	17313190	A1	Oceania
2025	TEA	AUSTRIA	KGS	10627.00	75751	A1	Europe
2025	TEA	AZERBAIJAN	KGS	624168.00	2152512	A1	Asia
2025	TEA	BAHARAIN IS	KGS	962855.00	4387382	A1	Asia
2025	TEA	BANGLADESH ...	KGS	445270.00	1875445	A1	Asia
2025	TEA	BARBADOS	KGS	251.00	11282	A1	North America
2025	TEA	BELARUS	KGS	750.00	15354	A1	Europe
2025	TEA	BELGIUM	KGS	83251.00	719995	A1	Europe
2025	TEA	BELIZE	KGS	255.00	1651	A1	North America
2025	TEA	BENIN	KGS	283467.00	1719746	A1	Africa
2025	TEA	BHUTAN	KGS	423381.00	1927597	A1	Asia
2025	TEA	BOLIVIA	KGS	13601.00	86289	A1	South America
2025	TEA	BOTSWANA	KGS	300.00	3552	A1	Africa

SQL Analysis

FY2024-25	FY2016-FY2026	FY2025 STATE-WISE
• Top 15 Export Countries • Top 15 Export Commodities • Top 5 Export Continents • Top Export Commodity for each Country • Top Export Commodity for each Continent • Largest Importer in every Continent • Largest Importer for every Commodity • Share of Top 20% Countries in Total Export	• CAGR • Country-wise CAGR • Commodity Share • Continent Share • Country Share • Year-wise Trade • Top 20 Countries • Top 20 Commodities • Top commodity in each continent • Top importer of each commodity • Top importing country in each continent	• Highest Exporting state for each commodity. • No. of commodities each state exports • No. of countries each state export to • Top 3 commodities of every state • Top 20 State-Country Pair • Top 20 States • Top commodity of each state • Top destination of each state

SQL Analysis

Top 15 Export Countries

	Rank	Country	Million USD
▶	1	UNITED STATES	86514
	2	U ARAB EMTS	36638
	3	NETHERLAND	22763
	4	U K	14550
	5	CHINA P RP	14252
	6	SINGAPORE	12976
	7	SAUDI ARAB	11757
	8	BANGLADESH PR	11485
	9	GERMANY	10629
	10	AUSTRALIA	8581
	11	FRANCE	7960
	12	ITALY	7727
	13	SOUTH AFRICA	7462
	14	NEPAL	7386
	15	MALAYSIA	7317

Top Export Commodity for each Continent

CONTINENT	PC_CODE	COMMODITY	Million USD
Asia	S6	PETROLEUM PRODUCTS	23810
Europe	S6	PETROLEUM PRODUCTS	16431
North America	P4	TELECOM INSTRUMENTS	11375
Africa	S6	PETROLEUM PRODUCTS	9956
Oceania	S6	PETROLEUM PRODUCTS	4670
South America	H9	AGRO CHEMICALS	1347
Antarctica	J5	CERAMICS AND ALLIED PRODUCTS	0

Largest Importer in every Continent

	CONTINENT	COUNTRY	Million USD
▶	North America	UNITED STATES	86514
	Asia	U ARAB EMTS	36638
	Europe	NETHERLAND	22763
	Oceania	AUSTRALIA	8581
	South America	BRAZIL	6771
	Africa	TANZANIA REP	4790

SQL Analysis

Largest Importer for every Commodity

	PC_CODE	COMMODITY	COUNTRY	Million USD
▶	S6	PETROLEUM PRODUCTS	NETHERLAND	13223
	P4	TELECOM INSTRUMENTS	UNITED STATES	10872
	H8	DRUG FORMULATIONS, BIOLOGICALS	UNITED STATES	9696
	G5	PEARL, PRECS, SEMIPRECS STONES	UNITED STATES	5717
	G9	GOLD AND OTH PRECS METL JWLRY	U ARAB EMTS	5485
	Q6	RMG COTTON INCL ACCESSORIES	UNITED STATES	3176
	L4	PRODUCTS OF IRON AND STEEL	UNITED STATES	3105
	P8	COTTON FABRICS, MADEUPS ETC.	UNITED STATES	2734
	N4	ELECTRIC MACHINERY AND EQUIPME	UNITED STATES	2723
	E7	MARINE PRODUCTS	UNITED STATES	2681
	P2	ELECTRONICS COMPONENTS	UNITED STATES	2035

Top 5 Continents

	Rank	Continent	Million USD
▶	1	Asia	177043
	2	North America	99330
	3	Europe	93041
	4	Africa	42747
	5	South America	12477

Top Commodity for each Country

COUNTRY	COMMODITY	Million USD
NETHERLAND	PETROLEUM PRODUCTS	13223
UNITED STATES	TELECOM INSTRUMENTS	10872
U ARAB EMTS	PETROLEUM PRODUCTS	6395
SINGAPORE	PETROLEUM PRODUCTS	4975
AUSTRALIA	PETROLEUM PRODUCTS	4486
HONG KONG	PEARL, PRECS, SEMIPRECS STONES	3685
UNSPECIFIED	PETROLEUM PRODUCTS	3243
TANZANIA REP	PETROLEUM PRODUCTS	2867
SOUTH AFRICA	PETROLEUM PRODUCTS	2819
MALAYSIA	PETROLEUM PRODUCTS	2274
NEPAL	PETROLEUM PRODUCTS	2088
CHINA P RP	IRON ORE	1885
TOGO	PETROLEUM PRODUCTS	1846
BANGLADESH PR	COTTON YARN	1749
BELGIUM	PEARL, PRECS, SEMIPRECS STONES	1533
SAUDI ARAB	MOTOR VEHICLE/CARS	1533
U K	TELECOM INSTRUMENTS	1515
OMAN	PETROLEUM PRODUCTS	1428
ITALY	TELECOM INSTRUMENTS	1263

SQL Analysis

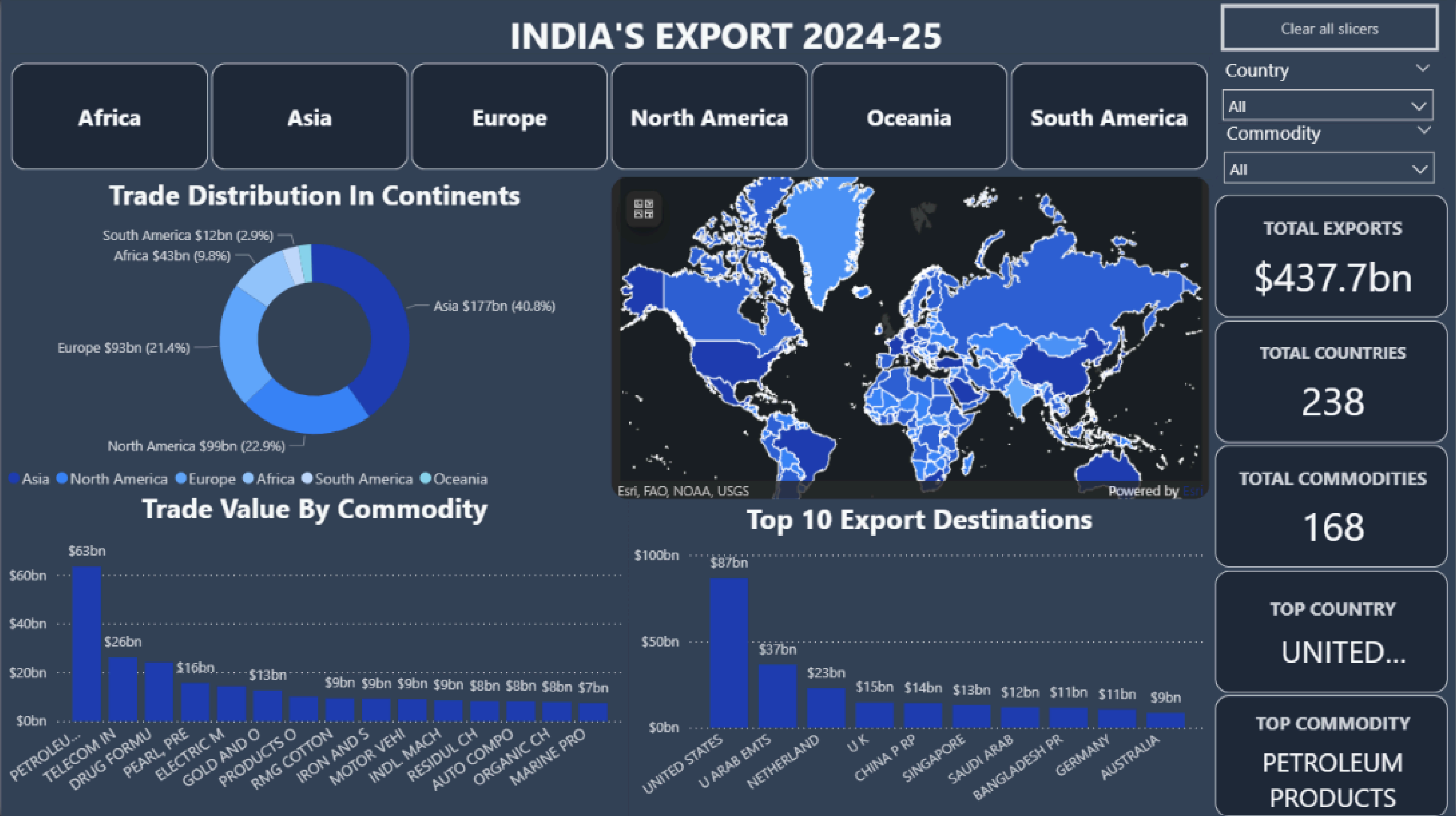
Top 15 Export Commodities

Rank	PC_Code	Commodity	Million USD
1	S6	PETROLEUM PRODUCTS	63383
2	P4	TELECOM INSTRUMENTS	26088
3	H8	DRUG FORMULATIONS, BIOLOGICALS	24146
4	G5	PEARL, PRECS, SEMIPRECS STONES	15588
5	N4	ELECTRIC MACHINERY AND EQUIPME	14380
6	G9	GOLD AND OTH PRECS METL JWLERY	12622
7	L4	PRODUCTS OF IRON AND STEEL	10117
8	Q6	RMG COTTON INCL ACCESSORIES	9381
9	L3	IRON AND STEEL	9251
10	O5	MOTOR VEHICLE/CARS	9004
11	N6	INDL. MACHNRY FOR DAIRY ETC	8507
12	I7	RESIDUL CHEMICL AND ALLED PROD	8222
13	M3	AUTO COMPONENTS/PARTS	8185
14	I3	ORGANIC CHEMICALS	7864
15	E7	MARINE PRODUCTS	7405

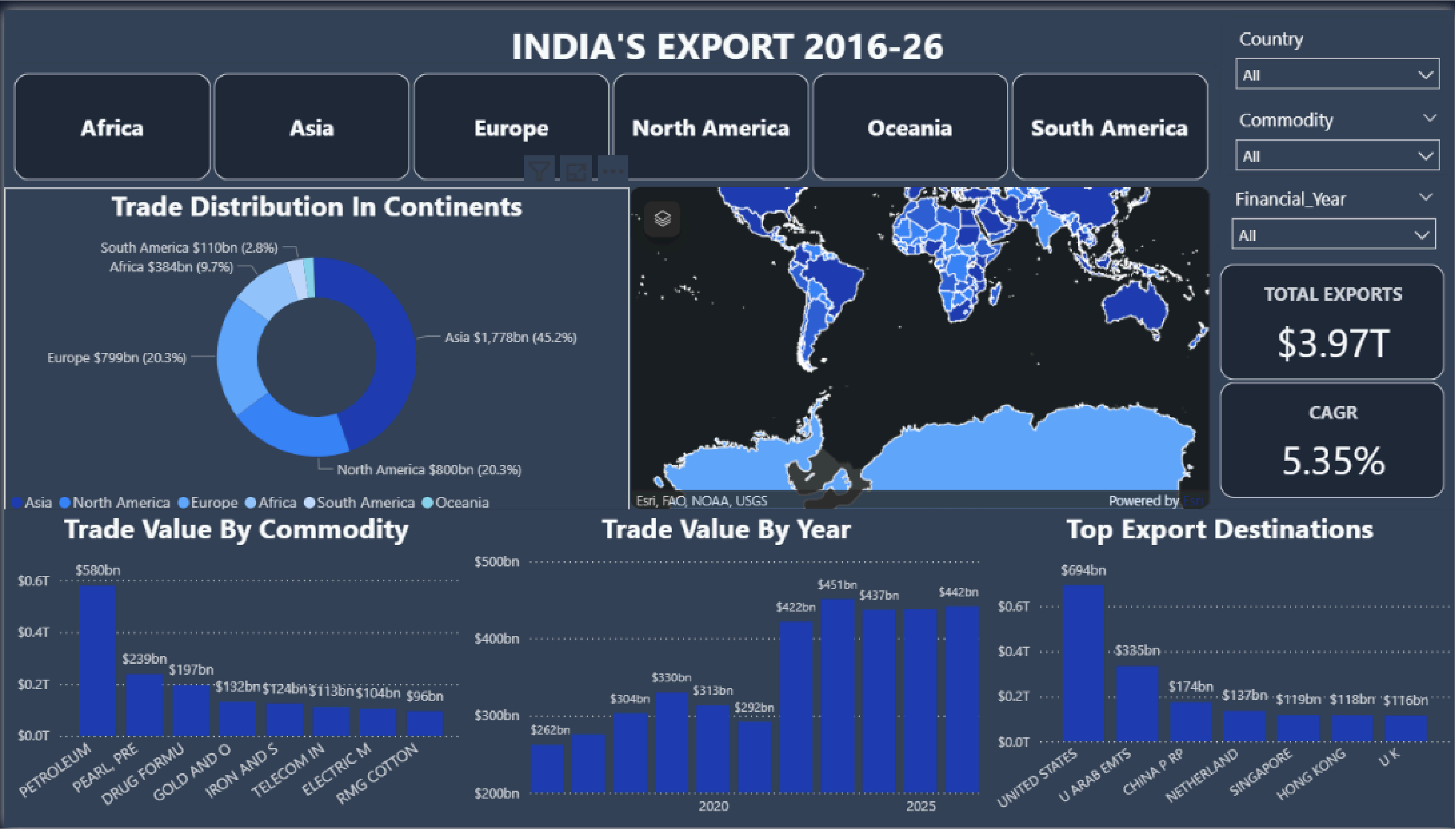
Power BI Dashboard Development

- KPI Cards
- Interactive World Map
- Top Export Countries
- Top Commodities
- Continent Distribution
- Country Filters
- Commodity Filters
- Continent Filters
- Total Export Value
- CAGR Metric
- Top Country Card
- Top Commodity Card

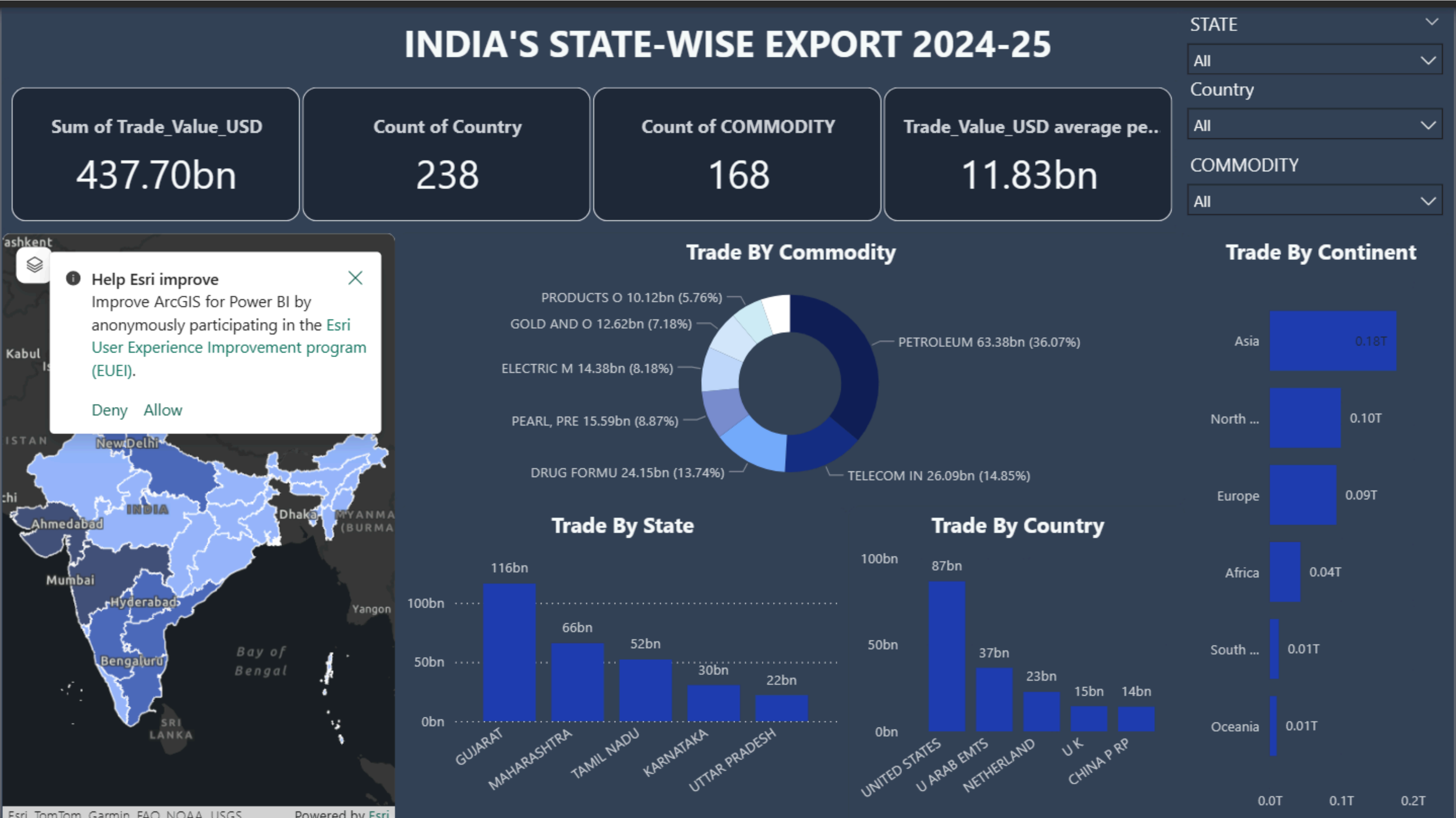
Power BI Dashboard FY-2025



Power BI Dashboard FY-2016 to FY-2026



Power BI Dashboard FY2025-Statewise



Insights — Current Findings

Overall trend

- Exports grew from \$262B in FY16 to \$442B in FY26, which works out to about **5.35% CAGR**.
- There was a big jump in **FY22 (+44.6%**, probably post-covid recovery + high oil prices)
- After that it's basically **flat: +6.9%, -3.1%, +0.1%, +0.9%** for the last four years.
- On paper exports look like they're growing, but almost all of that growth happened in one year and things have plateaued since.

Product concentration

- **Petroleum products** is the single largest commodity at **14.6%** of total exports (\$579B over the period).
- **Top 15 commodities** together **make up ~54%** of all exports, so there's a reasonable spread.
- The #1 category is tied to global oil prices rather than actual demand for Indian goods.

Insights — Current Findings

Country concentration

- **USA alone is 17.5%** of total exports (\$694B).
- Top 10 countries account for just over 50% of everything.
- US tariffs are at 10-27% (110% on patented pharma), and pharma is our 3rd biggest commodity with the US as its top buyer (\$73B). So, the biggest customer and the biggest tariff risk are the same country, and it overlaps with the biggest export category too.

Growing vs declining markets (min \$100M base in FY16)

- **Fastest growing:** Djibouti, Togo, Libya (~17% CAGR each), then Austria, Romania, Netherlands (~14%).
- **Declining:** Pakistan (-19%), Korea DPR (-28%), Gibraltar (-49%). These are mostly geopolitical, not really something to act on.

Insights — Current Findings

State level (FY25)

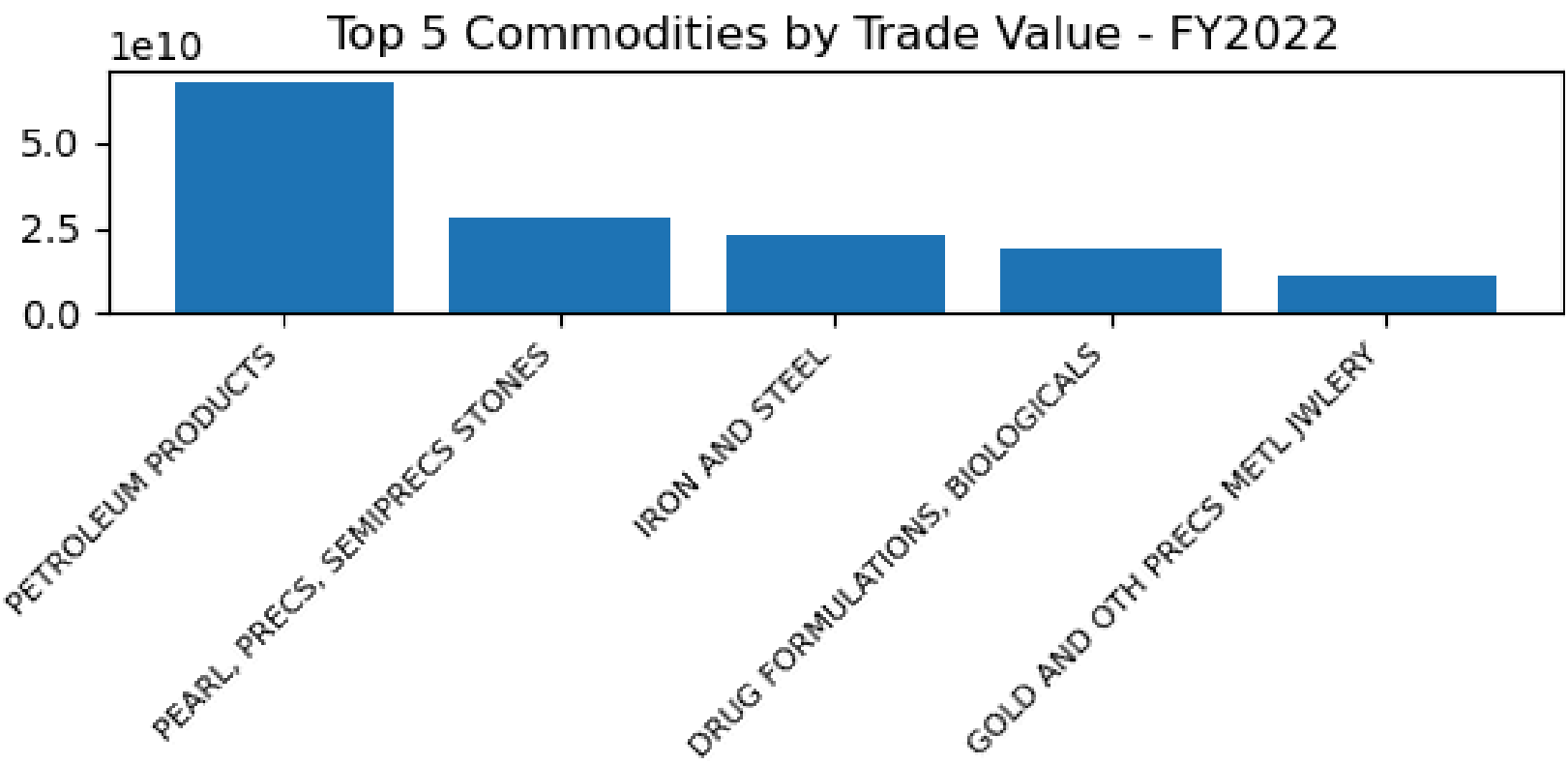
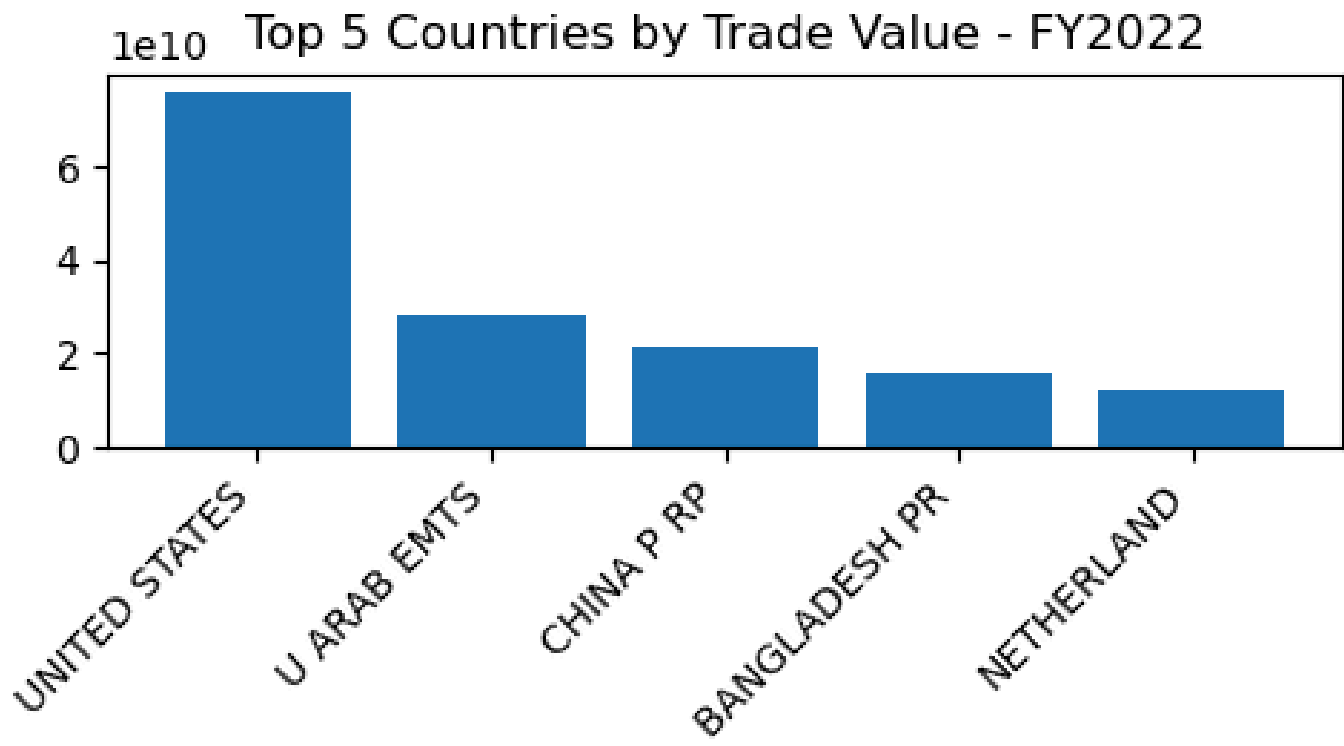
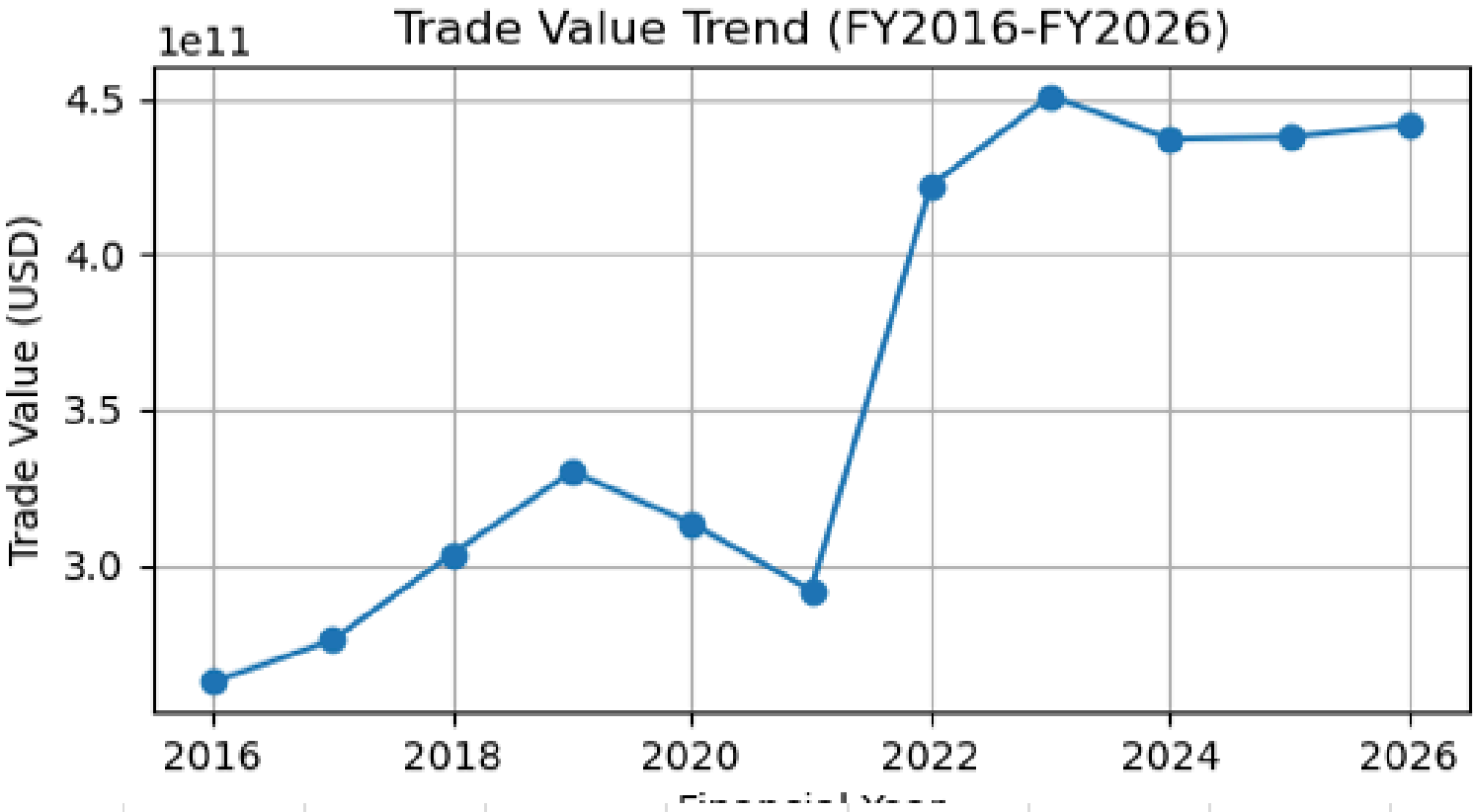
- **Gujarat leads with 26.6%** of state-wise exports.
- 38% of Gujarat's own basket is petroleum - so it's basically one industry carrying the state.
- Maharashtra (15%) and Tamil Nadu (12%) are next.
- **Top 3 states = ~54% of exports**, so the industrial base is pretty concentrated geographically.

Diversification

- UAE, USA, Nepal and Singapore each import all 168 tracked commodities from India - these are the most "complete" trade relationships and probably the safest markets to push new products into.
- RMG Cotton and "Other Misc Engineering Items" reach 232 countries each, the widest spread of any commodity.

Automated Yearly Trade Report

Export Performance Report - FY2022		
Total Trade Value (Billion USD)		422
Countries Traded With		230
Commodities Exported		168
Top Commodity	PETROLEUM PRODUCTS	
Top Country	UNITED STATES	
Top Continent	Asia	
Previous FY Trade Value (Billion USD)		291.81
YoY Change (%)	44.61%	



Trade Intelligence Resources Developed

Export Tariff Database — Top 50 Export Destination Countries

Country / Region	Available Modes	Primary Routes	Avg Transit Time (Days)
AUSTRALIA	Sea, Air	Sea via Melbourne / Sydney / Fremantle	18-24
AUSTRIA	Sea, Rail, Air	Sea + Rail via Hamburg	28-35
BANGLADESH PR	Sea, Road, Rail, Air	Sea via Chittagong	5-7
BELGIUM	Sea, Air	Sea via Antwerp	22-28
BHUTAN	Road, Air	Land via Phuentsholing	2-4
BRAZIL	Sea, Air	Sea via Santos / Itajai	28-35
CANADA	Sea, Air	Sea via Vancouver / Montreal / Halifax	25-35
CHINA P RP	Sea, Rail, Air	Sea via Shanghai/Tianjin	16-20
COLOMBIA	Sea, Air	Sea via Cartagena / Buenaventura	26-33
CZECH REPUBLIC	Sea, Rail, Air	Sea + Rail via Hamburg	26-33
EGYPT A RP	Sea, Air	Sea via Alexandria / Port Said	14-18
FRANCE	Sea, Air	Sea via Le Havre / Marseille	22-28
GERMANY	Sea, Air	Sea via Hamburg / Bremerhaven	22-28
GHANA	Sea, Air	Sea via Tema	22-26
HONG KONG	Sea, Air	Sea via Hong Kong	14-18
INDONESIA	Sea, Air	Sea via Jakarta (Tanjung Priok)	14-18
IRAQ	Sea, Air	Sea via Umm Qasr	12-16
ISRAEL	Sea, Air	Sea via Haifa (via Suez)	18-22
ITALY	Sea, Air	Sea via Genoa / Gioia Tauro (via Suez)	20-25
JAPAN	Sea, Air	Sea via Yokohama/Osaka	20-25
KENYA	Sea, Air	Sea via Mombasa	14-18
KOREA RP	Sea, Air	Sea via Busan	18-22
KUWAIT	Sea, Air	Sea via Shuwaikh	8-12

Country-wise Export Transport Mode and Trade Route Reference Data

RANK	COUNTRY	TRADE VALUE(MILLION USD)	TARRIFS	NOTES
1	UNITED STATES	86514	10%-27%	110% on Patented Pharma
2	U ARAB EMTS	36638	0	FTA
3	NETHERLAND	22763	4-6%	EU MFN
4	U K	14550	0	FTA
5	CHINA P RP	14252	7-8%	MFN
6	SINGAPORE	12976	0	FTA
7	SAUDI ARAB	11757	5%	GCC Tarrif
8	BANGLADESH PR	11485	0	FTA
9	GERMANY	10629	4-6%	EU MFN
10	AUSTRALIA	8581	0	FTA
11	FRANCE	7960	4-6%	EU MFN
12	ITALY	7727	4-6%	EU MFN
13	SOUTH AFRICA	7462	0	FTA
14	NEPAL	7386	0	FTA
15	MALAYSIA	7317	0	FTA
16	BRAZIL	6771	PTA	
17	BELGIUM	6320	4-6%	EU MFN
18	JAPAN	6249	0	FTA
19	HONG KONG	6067	0	NO TARRIFS FOR ANYONE
20	KOREA RP	5820	0	FTA
21	MEXICO	5749	7-8%	
22	TURKEY	5717	8-10%	
23	VIETNAM SOC REP	5429	0	FTA

Trade Intelligence Resources Developed

Top exporting Company Commodity wise

COMMODITY	COMPANY WITH HIGHEST TRADE	HEADQUARTERS
VEGETABLE OILS	ADANI WILMAR	AHEMDABAD, GUJARAT
RAW FABRIC TEXTILE	ARVIND MILLS	AHEMDABAD, GUJARAT
DAIRY PRODUCTS	AMUL	ANAND, GUJARAT
ELECTRONIC INSTRUMENTS	BHARAT ELECTRONICS LTD	BENGALURU, KARNATAKA
AIRCRAFT, SPACECRAFT ETC	HAL	BENGALURU, KARNATAKA
GOLD AND PRECIOUS METAL JEWELLERY	RAJESH EXPORTS	BENGALURU, KARNATAKA
ELECTRONICS COMPONENTS	TATA ELECTRONICS SYSTEMS SOLUTIO	BENGALURU, KARNATAKA
TELECOM INSTRUMENTS	FOXCONN HON HAI	CHENNAI, TAMIL NADU
AUTO TYRES	MRF	CHENNAI, TAMIL NADU
CONSTRUCTION MACHINERY	JCB INDIA	FARIDABAD, HARYANA
READY MADE GARMENTS	SHAHI EXPORTS	FARIDABAD, HARYANA
AYUSH AND HERBAL PRODUCTS	DABUR	GHAZIABAD, UTTAR PRADESH
DRUG INTERMEDIATES	DR. REDDY LAB.	HYDERABAD, TELENGANA
IRON ORE	NMDC	HYDERABAD, TELENGANA
SHIP,BOAT ETC	COCHIN SHIPYARD	KOCHI, KERALA
PLYWOOD	CENTURY PLYBOARDS	KOLKATA, WEST BENGAL
COAL COKE ETC	COAL INDIA LTD	KOLKATA, WEST BENGAL
TOBACCO UNMANUFACTURED	ITC	KOLKATA, WEST BENGAL
CEREALS	ITC	KOLKATA, WEST BENGAL
TOBACCO MANUFACTURED	ITC	KOLKATA, WEST BENGAL
MILLED PRODUCTS	ITC	KOLKATA, WEST BENGAL
TEA	MCLEOD RUSSEL INDIA	KOLKATA, WEST BENGAL
BICYCLE AND PARTS	HERO CYCLES LTD	LUDHIANA, PUNJAB
PAINT	ASIAN PAINTS	MUMBAI,MAHARASHTRA
GLASS AND GLASSWARE	BOROSIL RENEWABLES	MUMBAI,MAHARASHTRA
SPICES	EVEREST	MUMBAI,MAHARASHTRA

Compliance Guide

India Export Compliance Guide

Part 1 — Food & Agricultural Products

Country	Key Regulator(s)	Core Requirements
USA	FDA (FSVP), USDA	- FDA Food Facility Registration (every 2 yrs) - Foreign Supplier Verification Program (FSVP) compliance by US importer - USDA/APHIS approval for meat, dairy, plants - FDA labelling (Nutrition Facts format)
UAE	ESMA, MOIAT	- Halal certification - Product registration on UAE.gov/MOIAT portal - Arabic labelling
Netherlands And Germany (EU)	EU food law via RASFF, NVWA	- EU Health Certificate - compliance with EU food contaminant/pesticide MRLs - Novel Food approval if applicable
UK	FSA	- EU-style Health Certificates - UK-specific labelling (allergens, metric units)
China PRP	GACC	- Mandatory overseas manufacturer registration under GACC Decree No. 280 - Chinese-language labels with 18-digit GACC reg. number - Customs declaration must state registration number + “edible” use field
Singapore	SFA (Singapore Food Agency)	- SFA import license for importer - For high-risk foods (meat, eggs, fish, fresh produce) — accredited source country/establishment listing required - English labelling

Challenges Faced

- Collecting data spread across multiple Excel files for ten financial years
- Standardizing inconsistent country names across different source files
- Mapping countries to continents and handling regional classifications
- Cleaning and validating large export datasets for accuracy
- Designing meaningful and user-friendly Power BI visuals
- Preparing comprehensive trade intelligence reference datasets

Conclusion

The project has successfully progressed from raw data collection to the development of a comprehensive export analytics platform. A centralized export database has been created using DGCI&S data covering FY2016–17 to FY2025–26. Analytical SQL queries and an interactive Power BI dashboard have been developed to study export performance across commodities, countries, and continents.

Additionally, several trade intelligence resources — including export tariff datasets, Compliance guide, exporting companies and transport mode lookups — have been compiled to enhance the project's practical value.

Thank You